Class 10 Mathematics (NCERT) – Chapter 7: Coordinate Geometry

Worked solutions: Examples 1–10, Exercise 7.1 and Exercise 7.2

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Formulas used in this chapter

(F1) **Distance formula.** The distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

(F2) **Distance from the origin.** $OP = \sqrt{x^2 + y^2}$ for $P(x, y)$.

(F3) **Section formula (internal division).** The point P dividing the segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m_1 : m_2$ is $P\left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2},\ \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$.

(F4) **Mid-point formula.** The mid-point of $A(x_1, y_1)$ and $B(x_2, y_2)$ is $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$ (section formula with $m_1 : m_2 = 1 : 1$).

(F5) A point on the x-axis has the form $(x, 0)$ and a point on the y-axis has the form $(0, y)$.

Part A: Solved Examples 1 to 5 (Section 7.2, Distance Formula)

Example 1

Question. Do the points $(3, 2)$, $(-2, -3)$ and $(2, 3)$ form a triangle? If so, name the type of triangle formed.

Solution. Let $P(3, 2)$, $Q(-2, -3)$ and $R(2, 3)$.

Step 1: find the three sides. By the distance formula, the distance between (x_1, y_1) and (x_2, y_2) is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

How to read the working below: in each pair the first point named is (x_1, y_1) and the second is (x_2, y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the brackets, and then the squares are added and the square root is taken.

$$\begin{aligned}
PQ &= \sqrt{(-2 - 3)^2 + (-3 - 2)^2} \\
&= \sqrt{(-5)^2 + (-5)^2} \\
&= \sqrt{25 + 25} = \sqrt{50} = 5\sqrt{2} \approx 7.07 \\
QR &= \sqrt{(2 - (-2))^2 + (3 - (-3))^2} \\
&= \sqrt{4^2 + 6^2} \\
&= \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13} \approx 7.21 \\
PR &= \sqrt{(2 - 3)^2 + (3 - 2)^2} \\
&= \sqrt{(-1)^2 + 1^2} \\
&= \sqrt{1 + 1} = \sqrt{2} \approx 1.41
\end{aligned}$$

Common mistake. *A negative number squared is positive:* $(-5)^2 = 25$*, not* -25*. Also* $-2 - 3 = -5$ *(not* 1*), and* $2 - (-2) = 2 + 2 = 4$*.*

Step 2: do the points form a triangle? Three points form a triangle only if the sum of any two of the distances is greater than the third. Using the values above ($PQ \approx 7.07$, $QR \approx 7.21$, $PR \approx 1.41$):

$$PQ + PR \approx 7.07 + 1.41 = 8.48 > QR \ (\approx 7.21)$$
$$QR + PR \approx 7.21 + 1.41 = 8.62 > PQ \ (\approx 7.07)$$
$$PQ + QR \approx 7.07 + 7.21 = 14.28 > PR \ (\approx 1.41)$$

All three hold, so P, Q, R are not in a straight line and form a triangle.

Step 3: what type of triangle? By the converse of the Pythagoras theorem (if $a^2 + b^2 = c^2$ for the sides of a triangle, the angle opposite c is $90°$). The square of each side is the number under its square root in Step 1: $PQ^2 = 50$, $PR^2 = 2$, $QR^2 = 52$. The longest side is QR, so test it:

$$PQ^2 + PR^2 = 50 + 2 = 52 = QR^2.$$

So the angle opposite QR, which is $\angle P$, is $90°$. The sides PQ, QR, PR are all different, so the triangle is not isosceles; it is a right-angled triangle.

Answer: The points form a triangle. It is a right-angled triangle, with the right angle at P.

Example 2

Question. Show that the points $(1, 7)$, $(4, 2)$, $(-1, -1)$ and $(-4, 4)$ are the vertices of a square.

Solution. Let $A(1, 7)$, $B(4, 2)$, $C(-1, -1)$, $D(-4, 4)$.

A quadrilateral is a square if all four sides are equal and both diagonals are equal. By the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, find the four sides and the two diagonals.

How to read the working below: in each pair the first point named is (x_1, y_1) and the second is (x_2, y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the

brackets, and then the squares are added and the square root is taken.

$$\begin{aligned}
AB &= \sqrt{(4-1)^2 + (2-7)^2} \\
&= \sqrt{3^2 + (-5)^2} \\
&= \sqrt{9+25} = \sqrt{34} \\
BC &= \sqrt{(-1-4)^2 + (-1-2)^2} \\
&= \sqrt{(-5)^2 + (-3)^2} \\
&= \sqrt{25+9} = \sqrt{34} \\
CD &= \sqrt{(-4-(-1))^2 + (4-(-1))^2} \\
&= \sqrt{(-3)^2 + 5^2} \\
&= \sqrt{9+25} = \sqrt{34} \\
DA &= \sqrt{(1-(-4))^2 + (7-4)^2} \\
&= \sqrt{5^2 + 3^2} \\
&= \sqrt{25+9} = \sqrt{34} \\
AC &= \sqrt{(-1-1)^2 + (-1-7)^2} \\
&= \sqrt{(-2)^2 + (-8)^2} \\
&= \sqrt{4+64} = \sqrt{68} = 2\sqrt{17} \\
BD &= \sqrt{(-4-4)^2 + (4-2)^2} \\
&= \sqrt{(-8)^2 + 2^2} \\
&= \sqrt{64+4} = \sqrt{68} = 2\sqrt{17}
\end{aligned}$$

Here $\sqrt{34}$ cannot be simplified, and $\sqrt{68} = \sqrt{4 \times 17} = 2\sqrt{17}$.

Since $AB = BC = CD = DA = \sqrt{34}$ and $AC = BD = 2\sqrt{17}$, all four sides are equal and the two diagonals are equal.

Answer: $ABCD$ is a square.

Alternative solution. Using the converse of the Pythagoras theorem in $\triangle ADC$: $AD^2 + DC^2 = 34 + 34 = 68 = AC^2$ (the values 34, 34, 68 are the numbers under the square roots above), so $\angle D = 90°$. A quadrilateral with all four sides equal and one angle 90° is a square.

Example 3

Question. Fig. 7.6 shows the arrangement of desks in a classroom. Ashima, Bharti and Camella are seated at $A(3, 1)$, $B(6, 4)$ and $C(8, 6)$ respectively. Do you think they are seated in a line? Give reasons for your answer.

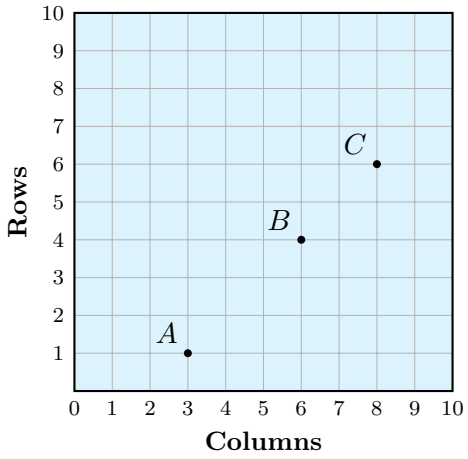


Fig. 7.6 (redrawn)

Solution. The coordinates of the seats are given in the question: $A(3,1)$, $B(6,4)$, $C(8,6)$.

Three points are collinear if the largest of the three distances equals the sum of the other two. By the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$:

How to read the working below: in each pair the first point named is (x_1, y_1) and the second is (x_2, y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the brackets, and then the squares are added and the square root is taken.

$$\begin{aligned} AB &= \sqrt{(6-3)^2 + (4-1)^2} \\ &= \sqrt{3^2 + 3^2} \\ &= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \\ BC &= \sqrt{(8-6)^2 + (6-4)^2} \\ &= \sqrt{2^2 + 2^2} \\ &= \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \\ AC &= \sqrt{(8-3)^2 + (6-1)^2} \\ &= \sqrt{5^2 + 5^2} \\ &= \sqrt{25+25} = \sqrt{50} = 5\sqrt{2} \end{aligned}$$

Here $\sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}$, $\sqrt{8} = \sqrt{4 \times 2} = 2\sqrt{2}$ and $\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$.

The largest distance is $AC = 5\sqrt{2}$. Add the other two:

$$AB + BC = 3\sqrt{2} + 2\sqrt{2} = (3+2)\sqrt{2} = 5\sqrt{2} = AC.$$

Answer: Since $AB + BC = AC$, the points A, B, C are collinear. Ashima, Bharti and Camella are seated in a line.

Example 4

Question. Find a relation between x and y such that the point (x, y) is equidistant from the points $(7, 1)$ and $(3, 5)$.

Solution. Let $P(x, y)$ be equidistant from $A(7, 1)$ and $B(3, 5)$, so $AP = BP$ and therefore $AP^2 = BP^2$ (squaring removes the square roots).

By the distance formula, $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$, so

$$AP^2 = (x - 7)^2 + (y - 1)^2, \qquad BP^2 = (x - 3)^2 + (y - 5)^2.$$

Equate them and expand each bracket with $(a - b)^2 = a^2 - 2ab + b^2$:

$$\begin{aligned}(x - 7)^2 + (y - 1)^2 &= (x - 3)^2 + (y - 5)^2\\ (x^2 - 14x + 49) + (y^2 - 2y + 1) &= (x^2 - 6x + 9) + (y^2 - 10y + 25)\\ x^2 + y^2 - 14x - 2y + 50 &= x^2 + y^2 - 6x - 10y + 34\end{aligned}$$

Cancel $x^2 + y^2$ from both sides and collect the terms on the left:

$$\begin{aligned}-14x - 2y + 50 &= -6x - 10y + 34\\ -14x + 6x - 2y + 10y + 50 - 34 &= 0\\ -8x + 8y + 16 &= 0\\ x - y &= 2 \qquad \text{(dividing by } -8)\end{aligned}$$

Common mistake. *When expanding $(x - 7)^2$ the middle term $-14x$ is often forgotten; $(x - 7)^2 \neq x^2 - 49$.*

Verification. The point $(4, 2)$ satisfies $x - y = 4 - 2 = 2$. Its distances from $A(7, 1)$ and $B(3, 5)$ are

$$\begin{aligned}AP &= \sqrt{(4 - 7)^2 + (2 - 1)^2}\\ &= \sqrt{(-3)^2 + 1^2}\\ &= \sqrt{9 + 1} = \sqrt{10}\\ BP &= \sqrt{(4 - 3)^2 + (2 - 5)^2}\\ &= \sqrt{1^2 + (-3)^2}\\ &= \sqrt{1 + 9} = \sqrt{10}\end{aligned}$$

Both are $\sqrt{10}$, so the point is equidistant, as required.

Answer: The required relation is $x - y = 2$.

Example 5

Question. Find a point on the y-axis which is equidistant from the points $A(6, 5)$ and $B(-4, 3)$.

Solution. A point on the y-axis has the form $(0, y)$ (its x-coordinate is 0). Let $P(0, y)$ be equidistant from A and B, so $PA = PB$ and $PA^2 = PB^2$.

By the distance formula $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$:

$$PA^2 = (6 - 0)^2 + (5 - y)^2, \qquad PB^2 = (-4 - 0)^2 + (3 - y)^2.$$

Equate them and expand with $(a-b)^2 = a^2 - 2ab + b^2$:

$$\begin{aligned}
(6-0)^2 + (5-y)^2 &= (-4-0)^2 + (3-y)^2 \\
36 + (25 - 10y + y^2) &= 16 + (9 - 6y + y^2) \\
61 - 10y + y^2 &= 25 - 6y + y^2 \\
61 - 10y &= 25 - 6y \\
36 &= 4y \\
y &= 9
\end{aligned}$$

So $P = (0, 9)$.

Verification. With $P(0, 9)$:

$$\begin{aligned}
PA &= \sqrt{(6-0)^2 + (5-9)^2} \\
&= \sqrt{6^2 + (-4)^2} \\
&= \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13} \\
PB &= \sqrt{(-4-0)^2 + (3-9)^2} \\
&= \sqrt{(-4)^2 + (-6)^2} \\
&= \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}
\end{aligned}$$

Both distances are $\sqrt{52} = 2\sqrt{13}$.

Answer: The required point is $(0, 9)$.

Part B: Exercise 7.1

Exercise 7.1, Question 1

Question. Find the distance between the following pairs of points:
(i) $(2,3),\ (4,1)$
(ii) $(-5,7),\ (-1,3)$
(iii) $(a,b),\ (-a,-b)$

Solution. By the distance formula, the distance between $P(x_1,y_1)$ and $Q(x_2,y_2)$ is

$$PQ=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}.$$

How to read the working below: in each pair the first point named is (x_1,y_1) and the second is (x_2,y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the brackets, and then the squares are added and the square root is taken.

(i) $(x_1,y_1)=(2,3)$ and $(x_2,y_2)=(4,1)$:

$$\begin{aligned} d&=\sqrt{(4-2)^2+(1-3)^2}\\ &=\sqrt{2^2+(-2)^2}\\ &=\sqrt{4+4}=\sqrt{8}=2\sqrt{2} \end{aligned}$$

Here $\sqrt{8}=\sqrt{4\times 2}=2\sqrt{2}$.

(ii) $(x_1,y_1)=(-5,7)$ and $(x_2,y_2)=(-1,3)$:

$$\begin{aligned} d&=\sqrt{(-1-(-5))^2+(3-7)^2}\\ &=\sqrt{4^2+(-4)^2}\\ &=\sqrt{16+16}=\sqrt{32}=4\sqrt{2} \end{aligned}$$

Here $\sqrt{32}=\sqrt{16\times 2}=4\sqrt{2}$.

(iii) $(x_1,y_1)=(a,b)$ and $(x_2,y_2)=(-a,-b)$:

$$\begin{aligned} d&=\sqrt{(-a-a)^2+(-b-b)^2}\\ &=\sqrt{(-2a)^2+(-2b)^2}\\ &=\sqrt{4a^2+4b^2}=\sqrt{4(a^2+b^2)}=2\sqrt{a^2+b^2} \end{aligned}$$

Common mistake. *$\sqrt{4a^2+4b^2}$ is not $2a+2b$: the square root of a sum is not the sum of the square roots. Factor out 4 first.*

Verification. For (iii) take $a=3$, $b=4$. The points are $(3,4)$ and $(-3,-4)$:

$$\sqrt{(-3-3)^2+(-4-4)^2}=\sqrt{(-6)^2+(-8)^2}=\sqrt{36+64}=\sqrt{100}=10,$$

and the formula gives $2\sqrt{3^2+4^2}=2\sqrt{25}=2\times 5=10$. The two agree.

Answer: (i) $2\sqrt{2}$ units; (ii) $4\sqrt{2}$ units; (iii) $2\sqrt{a^2+b^2}$ units.

Exercise 7.1, Question 2

Question. Find the distance between the points $(0, 0)$ and $(36, 15)$. Can you now find the distance between the two towns A and B discussed in Section 7.2.

(Section 7.2: A town B is located 36 km east and 15 km north of the town A.)

Solution. By the distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, with $(x_1, y_1) = (0, 0)$ and $(x_2, y_2) = (36, 15)$:

$$\begin{aligned} d &= \sqrt{(36 - 0)^2 + (15 - 0)^2} \\ &= \sqrt{36^2 + 15^2} \\ &= \sqrt{1296 + 225} = \sqrt{1521} = 39 \end{aligned}$$

Here $36^2 = 1296$, $15^2 = 225$, $1296 + 225 = 1521$ and $39^2 = 1521$, so $\sqrt{1521} = 39$.

Now the towns. Take town A as the origin $(0, 0)$, one unit as 1 km, east as the positive x-direction and north as the positive y-direction. Then town B, 36 km east and 15 km north of A, is at $(36, 15)$. So the distance between the towns is the distance between $(0, 0)$ and $(36, 15)$.

Answer: The distance between $(0, 0)$ and $(36, 15)$ is 39 units. Yes, the distance between the towns A and B is 39 km.

Exercise 7.1, Question 3

Question. Determine if the points $(1, 5)$, $(2, 3)$ and $(-2, -11)$ are collinear.

Solution. Let $A(1, 5)$, $B(2, 3)$, $C(-2, -11)$.

Three points are collinear if the largest of the three distances equals the sum of the other two. By the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$:

How to read the working below: in each pair the first point named is (x_1, y_1) and the second is (x_2, y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the brackets, and then the squares are added and the square root is taken.

$$\begin{aligned} AB &= \sqrt{(2 - 1)^2 + (3 - 5)^2} \\ &= \sqrt{1^2 + (-2)^2} \\ &= \sqrt{1 + 4} = \sqrt{5} \approx 2.24 \\ BC &= \sqrt{(-2 - 2)^2 + (-11 - 3)^2} \\ &= \sqrt{(-4)^2 + (-14)^2} \\ &= \sqrt{16 + 196} = \sqrt{212} = 2\sqrt{53} \approx 14.56 \\ AC &= \sqrt{(-2 - 1)^2 + (-11 - 5)^2} \\ &= \sqrt{(-3)^2 + (-16)^2} \\ &= \sqrt{9 + 256} = \sqrt{265} \approx 16.28 \end{aligned}$$

Here $\sqrt{212} = \sqrt{4 \times 53} = 2\sqrt{53}$.

The largest distance is $AC \approx 16.28$. The sum of the other two is

$$AB + BC \approx 2.24 + 14.56 = 16.80 \neq 16.28 \approx AC.$$

For an exact check, suppose $\sqrt{5} + \sqrt{212} = \sqrt{265}$. Squaring both sides, with $(p+q)^2 = p^2 + 2pq + q^2$:

$$5 + 212 + 2\sqrt{5 \times 212} = 265 \;\Rightarrow\; 2\sqrt{1060} = 48 \;\Rightarrow\; \sqrt{1060} = 24 \;\Rightarrow\; 1060 = 24^2 = 576,$$

which is false. So $AB + BC \neq AC$.

Common mistake. *Compare the largest distance with the sum of the* other two. *Comparing $AB + AC$ with BC proves nothing.*

Alternative solution (section formula). If C lay on line AB, it would divide AB in some ratio $k : 1$, and by the section formula $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$ with $A(1, 5)$, $B(2, 3)$, $m_1 : m_2 = k : 1$,

$$C = \left(\frac{k(2) + 1(1)}{k + 1}, \frac{k(3) + 1(5)}{k + 1}\right) = \left(\frac{2k + 1}{k + 1}, \frac{3k + 5}{k + 1}\right).$$

From the x-coordinate: $\frac{2k + 1}{k + 1} = -2 \Rightarrow 2k + 1 = -2k - 2 \Rightarrow k = -\frac{3}{4}$.
From the y-coordinate: $\frac{3k + 5}{k + 1} = -11 \Rightarrow 3k + 5 = -11k - 11 \Rightarrow k = -\frac{8}{7}$.
The two values of k differ, so no single ratio puts C on line AB.

Answer: The points are not collinear.

Exercise 7.1, Question 4

Question. Check whether $(5, -2)$, $(6, 4)$ and $(7, -2)$ are the vertices of an isosceles triangle.

Solution. Let $A(5, -2)$, $B(6, 4)$, $C(7, -2)$.

By the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$:

How to read the working below: in each pair the first point named is (x_1, y_1) and the second is (x_2, y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the brackets, and then the squares are added and the square root is taken.

$$\begin{aligned}
AB &= \sqrt{(6 - 5)^2 + (4 - (-2))^2} \\
&= \sqrt{1^2 + 6^2} \\
&= \sqrt{1 + 36} = \sqrt{37} \\
BC &= \sqrt{(7 - 6)^2 + (-2 - 4)^2} \\
&= \sqrt{1^2 + (-6)^2} \\
&= \sqrt{1 + 36} = \sqrt{37} \\
AC &= \sqrt{(7 - 5)^2 + (-2 - (-2))^2} \\
&= \sqrt{2^2 + 0^2} \\
&= \sqrt{4 + 0} = \sqrt{4} = 2
\end{aligned}$$

(Here $\sqrt{37} \approx 6.08$, and $\sqrt{4} = 2$.)

Do the points form a triangle? They do if the sum of any two distances is greater than the third:

$$\begin{aligned}
AB + BC &\approx 6.08 + 6.08 = 12.16 > AC = 2 \\
AB + AC &\approx 6.08 + 2 = 8.08 > BC \approx 6.08 \\
BC + AC &\approx 6.08 + 2 = 8.08 > AB \approx 6.08
\end{aligned}$$

Is it isosceles? A triangle is isosceles if two of its sides are equal. Here $AB = BC = \sqrt{37}$.

Answer: Yes, the points are the vertices of an isosceles triangle, with $AB = BC = \sqrt{37}$ units.

Exercise 7.1, Question 5

Question. In a classroom, 4 friends are seated at the points A, B, C and D as shown in Fig. 7.8. Champa and Chameli walk into the class and after observing for a few minutes Champa asks Chameli, "Don't you think $ABCD$ is a square?" Chameli disagrees. Using distance formula, find which of them is correct.

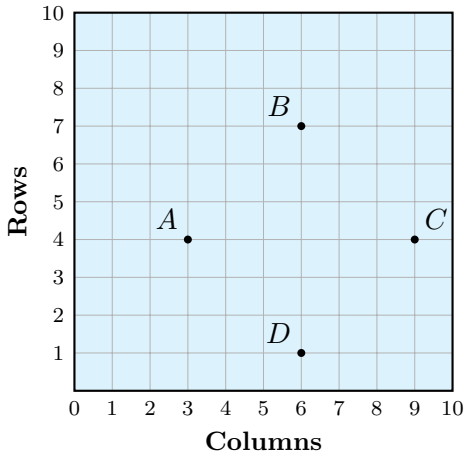


Fig. 7.8 (redrawn)

Solution. *Reading the figure.* In Fig. 7.8 the horizontal axis is labelled Columns and the vertical axis is labelled Rows. We take the column number as the x-coordinate and the row number as the y-coordinate. From the figure: A is in column 3, row 4, so $A(3,4)$; $B(6,7)$; $C(9,4)$; $D(6,1)$.

A quadrilateral is a square if all four sides are equal and both diagonals are equal. By the distance formula $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$:

How to read the working below: in each pair the first point named is (x_1, y_1) and the second is (x_2, y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the

brackets, and then the squares are added and the square root is taken.

$$\begin{aligned}
AB &= \sqrt{(6-3)^2+(7-4)^2} \\
&= \sqrt{3^2+3^2} \\
&= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \\
BC &= \sqrt{(9-6)^2+(4-7)^2} \\
&= \sqrt{3^2+(-3)^2} \\
&= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \\
CD &= \sqrt{(6-9)^2+(1-4)^2} \\
&= \sqrt{(-3)^2+(-3)^2} \\
&= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \\
DA &= \sqrt{(3-6)^2+(4-1)^2} \\
&= \sqrt{(-3)^2+3^2} \\
&= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \\
AC &= \sqrt{(9-3)^2+(4-4)^2} \\
&= \sqrt{6^2+0^2} \\
&= \sqrt{36+0} = \sqrt{36} = 6 \\
BD &= \sqrt{(6-6)^2+(1-7)^2} \\
&= \sqrt{0^2+(-6)^2} \\
&= \sqrt{0+36} = \sqrt{36} = 6
\end{aligned}$$

Here $\sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}$.

All four sides are equal ($3\sqrt{2}$ each) and the diagonals AC and BD are equal (6 each).

Common mistake. *Four equal sides alone also describe a rhombus. Always check the diagonals before calling it a square.*

Alternative solution. By the converse of the Pythagoras theorem, $AB^2 + BC^2 = 18 + 18 = 36 = AC^2$ (the numbers 18, 18 and 36 are those under the square roots above), so $\angle B = 90°$. A quadrilateral with four equal sides and one right angle is a square.

Answer: $ABCD$ is a square. Champa is correct and Chameli is wrong.

Exercise 7.1, Question 6

Question. Name the type of quadrilateral formed, if any, by the following points, and give reasons for your answer:
(i) $(-1,-2),\ (1,0),\ (-1,2),\ (-3,0)$
(ii) $(-3,5),\ (3,1),\ (0,3),\ (-1,-4)$
(iii) $(4,5),\ (7,6),\ (4,3),\ (1,2)$

Solution.

Method for every part.

1. Name the points A, B, C, D in the order given.
2. Use the distance formula $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ to find the four sides AB, BC, CD, DA and the two diagonals AC, BD.
3. Compare the lengths and read the type from the table.

What the lengths show	Type of quadrilateral
All four sides equal and diagonals equal	Square
All four sides equal but diagonals not equal	Rhombus
Opposite sides equal and diagonals equal (adjacent sides not equal)	Rectangle
Opposite sides equal but diagonals not equal	Parallelogram
Three of the four points lie on one straight line	No quadrilateral is formed

Common mistake. *Check the order of the points. The quadrilateral is joined $A \to B \to C \to D \to A$, so AC and BD are the diagonals.*

How to read the working below: in each pair the first point named is (x_1, y_1) and the second is (x_2, y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the brackets, and then the squares are added and the square root is taken.

(i) $A(-1,-2)$, $B(1,0)$, $C(-1,2)$, $D(-3,0)$.

$$\begin{aligned}
AB &= \sqrt{(1-(-1))^2+(0-(-2))^2} \\
&= \sqrt{2^2+2^2} \\
&= \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \\
BC &= \sqrt{(-1-1)^2+(2-0)^2} \\
&= \sqrt{(-2)^2+2^2} \\
&= \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \\
CD &= \sqrt{(-3-(-1))^2+(0-2)^2} \\
&= \sqrt{(-2)^2+(-2)^2} \\
&= \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \\
DA &= \sqrt{(-1-(-3))^2+(-2-0)^2} \\
&= \sqrt{2^2+(-2)^2} \\
&= \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \\
AC &= \sqrt{(-1-(-1))^2+(2-(-2))^2} \\
&= \sqrt{0^2+4^2} \\
&= \sqrt{0+16} = \sqrt{16} = 4 \\
BD &= \sqrt{(-3-1)^2+(0-0)^2} \\
&= \sqrt{(-4)^2+0^2} \\
&= \sqrt{16+0} = \sqrt{16} = 4
\end{aligned}$$

(Here $\sqrt{8} = 2\sqrt{2}$.) All four sides are equal ($2\sqrt{2}$) and the diagonals are equal (4). So $ABCD$ is a square.

(ii) $A(-3,5)$, $B(3,1)$, $C(0,3)$, $D(-1,-4)$.

$$\begin{aligned}
AB &= \sqrt{(3-(-3))^2+(1-5)^2} \\
&= \sqrt{6^2+(-4)^2} \\
&= \sqrt{36+16} = \sqrt{52} = 2\sqrt{13} \\
BC &= \sqrt{(0-3)^2+(3-1)^2} \\
&= \sqrt{(-3)^2+2^2} \\
&= \sqrt{9+4} = \sqrt{13} \\
AC &= \sqrt{(0-(-3))^2+(3-5)^2} \\
&= \sqrt{3^2+(-2)^2} \\
&= \sqrt{9+4} = \sqrt{13}
\end{aligned}$$

(Here $\sqrt{52} = \sqrt{4\times 13} = 2\sqrt{13}$.) Then $AC + CB = \sqrt{13} + \sqrt{13} = 2\sqrt{13} = AB$, so A, C, B are collinear.

Check with the mid-point formula $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$: the mid-point of AB is $\left(\frac{-3+3}{2}, \frac{5+1}{2}\right) = (0,3) = C$. So C lies on AB.

Among any four points, if three are collinear, they cannot be three corners of a quadrilateral, whatever the order of the vertices. So no quadrilateral is formed.

(iii) $A(4,5)$, $B(7,6)$, $C(4,3)$, $D(1,2)$.

$$\begin{aligned}
AB &= \sqrt{(7-4)^2+(6-5)^2} \\
&= \sqrt{3^2+1^2} \\
&= \sqrt{9+1} = \sqrt{10} \\
BC &= \sqrt{(4-7)^2+(3-6)^2} \\
&= \sqrt{(-3)^2+(-3)^2} \\
&= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \\
CD &= \sqrt{(1-4)^2+(2-3)^2} \\
&= \sqrt{(-3)^2+(-1)^2} \\
&= \sqrt{9+1} = \sqrt{10} \\
DA &= \sqrt{(4-1)^2+(5-2)^2} \\
&= \sqrt{3^2+3^2} \\
&= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2} \\
AC &= \sqrt{(4-4)^2+(3-5)^2} \\
&= \sqrt{0^2+(-2)^2} \\
&= \sqrt{0+4} = \sqrt{4} = 2 \\
BD &= \sqrt{(1-7)^2+(2-6)^2} \\
&= \sqrt{(-6)^2+(-4)^2} \\
&= \sqrt{36+16} = \sqrt{52} = 2\sqrt{13}
\end{aligned}$$

(Here $\sqrt{18} = 3\sqrt{2}$ and $\sqrt{52} = 2\sqrt{13}$.) Opposite sides are equal: $AB = CD = \sqrt{10}$ and $BC = DA = 3\sqrt{2}$. Adjacent sides are unequal ($\sqrt{10} \neq 3\sqrt{2}$), so it is not a rhombus or a square. The diagonals are unequal ($AC = 2$, $BD = 2\sqrt{13}$), so it is not a rectangle. So $ABCD$ is a parallelogram.

Verification for (iii). By the mid-point formula, the mid-point of AC is $\left(\frac{4+4}{2}, \frac{5+3}{2}\right) = (4,4)$ and the mid-point of BD is $\left(\frac{7+1}{2}, \frac{6+2}{2}\right) = (4,4)$. The diagonals bisect each other, which confirms a parallelogram.

Answer: (i) $ABCD$ is a square. (ii) No quadrilateral is formed (A, C, B are collinear). (iii) $ABCD$ is a parallelogram.

Exercise 7.1, Question 7

Question. Find the point on the x-axis which is equidistant from $(2,-5)$ and $(-2,9)$.

Solution. A point on the x-axis has the form $(x,0)$ (its y-coordinate is 0). Let $P(x,0)$, $A(2,-5)$, $B(-2,9)$. Since $PA = PB$, we have $PA^2 = PB^2$.

By the distance formula $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$:

$$PA^2 = (2-x)^2 + (-5-0)^2, \qquad PB^2 = (-2-x)^2 + (9-0)^2.$$

Since $(2-x)^2 = (x-2)^2$ and $(-2-x)^2 = (x+2)^2$, equate them and expand with $(a\pm b)^2 = a^2 \pm 2ab + b^2$:

$$\begin{aligned}
(x-2)^2 + (-5)^2 &= (x+2)^2 + 9^2 \\
(x^2 - 4x + 4) + 25 &= (x^2 + 4x + 4) + 81 \\
x^2 - 4x + 29 &= x^2 + 4x + 85 \\
-4x + 29 &= 4x + 85 \\
-8x &= 56 \\
x &= -7
\end{aligned}$$

So $P = (-7,0)$.

Common mistake. *The point is on the x-axis, so its y-coordinate is 0 (not its x-coordinate). Take care with the sign in $(x+2)^2$.*

Verification. With $P(-7,0)$:

$$\begin{aligned}
PA &= \sqrt{(2-(-7))^2 + (-5-0)^2} \\
&= \sqrt{9^2 + (-5)^2} \\
&= \sqrt{81+25} = \sqrt{106} \\
PB &= \sqrt{(-2-(-7))^2 + (9-0)^2} \\
&= \sqrt{5^2 + 9^2} \\
&= \sqrt{25+81} = \sqrt{106}
\end{aligned}$$

Both distances are $\sqrt{106}$.

Answer: The required point is $(-7,0)$.

Exercise 7.1, Question 8

Question. Find the values of y for which the distance between the points $P(2,-3)$ and $Q(10,y)$ is 10 units.

Solution. By the distance formula $PQ = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ with $(x_1,y_1) = (2,-3)$, $(x_2,y_2) = (10,y)$ and $PQ = 10$:

$$\begin{aligned}
\sqrt{(10-2)^2+(y-(-3))^2} &= 10 \\
\sqrt{8^2+(y+3)^2} &= 10 \\
64+(y+3)^2 &= 100 \qquad \text{(squaring both sides)} \\
(y+3)^2 &= 36 \\
y+3 &= \pm 6
\end{aligned}$$

So $y+3=6$ or $y+3=-6$, that is $y=3$ or $y=-9$.

Common mistake. *$(y+3)^2 = 36$ has two solutions, $+6$ and -6. Writing only $y+3=6$ loses the second point.*

Verification. For $y=3$, $Q=(10,3)$:

$$PQ = \sqrt{(10-2)^2+(3-(-3))^2} = \sqrt{8^2+6^2} = \sqrt{64+36} = \sqrt{100} = 10.$$

For $y=-9$, $Q=(10,-9)$:

$$PQ = \sqrt{(10-2)^2+(-9-(-3))^2} = \sqrt{8^2+(-6)^2} = \sqrt{64+36} = \sqrt{100} = 10.$$

Answer: $y=3$ or $y=-9$.

Exercise 7.1, Question 9

Question. If $Q(0,1)$ is equidistant from $P(5,-3)$ and $R(x,6)$, find the values of x. Also find the distances QR and PR.

Solution. Since Q is equidistant from P and R, $QP = QR$, so $QP^2 = QR^2$.

By the distance formula $d^2 = (x_2-x_1)^2+(y_2-y_1)^2$:

$$\begin{aligned}
QP^2 &= (5-0)^2+(-3-1)^2 = 5^2+(-4)^2 = 25+16 = 41 \\
QR^2 &= (x-0)^2+(6-1)^2 = x^2+5^2 = x^2+25
\end{aligned}$$

Equating, $x^2+25=41$, so $x^2=16$ and $x=4$ or $x=-4$.

The distances. $QR^2 = x^2+25 = 16+25 = 41$ for both values of x, so $QR = \sqrt{41}$.

For PR, take $(x_1,y_1)=(5,-3)$ and $(x_2,y_2)=(x,6)$: For $x=4$, $R=(4,6)$:

$$\begin{aligned}
PR &= \sqrt{(4-5)^2+(6-(-3))^2} \\
&= \sqrt{(-1)^2+9^2} \\
&= \sqrt{1+81} = \sqrt{82}
\end{aligned}$$

For $x=-4$, $R=(-4,6)$:

$$\begin{aligned}
PR &= \sqrt{(-4-5)^2+(6-(-3))^2} \\
&= \sqrt{(-9)^2+9^2} \\
&= \sqrt{81+81} = \sqrt{162} = 9\sqrt{2}
\end{aligned}$$

(Here $\sqrt{162} = \sqrt{81 \times 2} = 9\sqrt{2}$.)

Common mistake. *$x^2 = 16$ gives both $x = 4$ and $x = -4$, and the two points R give different values of PR.*

Answer: $x = 4$ or $x = -4$; $QR = \sqrt{41}$ units; $PR = \sqrt{82}$ units when $x = 4$ and $PR = 9\sqrt{2}$ units when $x = -4$.

Exercise 7.1, Question 10

Question. Find a relation between x and y such that the point (x, y) is equidistant from the point $(3, 6)$ and $(-3, 4)$.

Solution. Let $P(x, y)$ be equidistant from $A(3, 6)$ and $B(-3, 4)$, so $PA = PB$ and $PA^2 = PB^2$.

By the distance formula $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$:

$$PA^2 = (x-3)^2 + (y-6)^2, \qquad PB^2 = (x-(-3))^2 + (y-4)^2 = (x+3)^2 + (y-4)^2.$$

Equate them and expand with $(a \pm b)^2 = a^2 \pm 2ab + b^2$:

$$\begin{aligned}
(x-3)^2 + (y-6)^2 &= (x+3)^2 + (y-4)^2 \\
(x^2 - 6x + 9) + (y^2 - 12y + 36) &= (x^2 + 6x + 9) + (y^2 - 8y + 16) \\
x^2 + y^2 - 6x - 12y + 45 &= x^2 + y^2 + 6x - 8y + 25 \\
-6x - 12y + 45 &= 6x - 8y + 25 \\
45 - 25 &= 6x + 6x + 12y - 8y \\
20 &= 12x + 4y \\
3x + y &= 5 \qquad \text{(dividing by 4)}
\end{aligned}$$

Verification. The point $(1, 2)$ satisfies $3x + y = 3(1) + 2 = 5$. Its distances from $A(3, 6)$ and $B(-3, 4)$ are

$$\begin{aligned}
PA &= \sqrt{(1-3)^2 + (2-6)^2} \\
&= \sqrt{(-2)^2 + (-4)^2} \\
&= \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5} \\
PB &= \sqrt{(1-(-3))^2 + (2-4)^2} \\
&= \sqrt{4^2 + (-2)^2} \\
&= \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}
\end{aligned}$$

Both equal $\sqrt{20}$, so the point is equidistant.

Answer: The required relation is $3x + y = 5$.

Part C: Solved Examples 6 to 10 (Section 7.3, Section Formula)

Example 6

Question. Find the coordinates of the point which divides the line segment joining the points $(4, -3)$ and $(8, 5)$ in the ratio $3 : 1$ internally.

Solution. Let $A(4, -3)$, $B(8, 5)$ and let $P(x, y)$ divide AB internally in the ratio $m_1 : m_2 = 3 : 1$. So $(x_1, y_1) = (4, -3)$, $(x_2, y_2) = (8, 5)$, $m_1 = 3$, $m_2 = 1$.

By the section formula, $P\left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$:

$$x = \frac{3(8) + 1(4)}{3 + 1} = \frac{24 + 4}{4} = \frac{28}{4} = 7$$
$$y = \frac{3(5) + 1(-3)}{3 + 1} = \frac{15 - 3}{4} = \frac{12}{4} = 3$$

Common mistake. *m_1 multiplies the x_2 (and y_2) of the* far *point B, and m_2 multiplies the near point A. Swapping them gives the point that divides in the ratio $1 : 3$ instead.*

Answer: The required point is $(7, 3)$.

Example 7

Question. In what ratio does the point $(-4, 6)$ divide the line segment joining the points $A(-6, 10)$ and $B(3, -8)$?

Solution. Let $P(-4, 6)$ divide AB internally in the ratio $m_1 : m_2$. Here $(x_1, y_1) = (-6, 10)$ and $(x_2, y_2) = (3, -8)$.

By the section formula $P\left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$:

$$(-4, 6) = \left(\frac{m_1(3) + m_2(-6)}{m_1 + m_2}, \ \frac{m_1(-8) + m_2(10)}{m_1 + m_2}\right) = \left(\frac{3m_1 - 6m_2}{m_1 + m_2}, \ \frac{-8m_1 + 10m_2}{m_1 + m_2}\right).$$

Equating the x-coordinates:

$$\frac{3m_1 - 6m_2}{m_1 + m_2} = -4$$
$$3m_1 - 6m_2 = -4(m_1 + m_2) = -4m_1 - 4m_2$$
$$3m_1 + 4m_1 = 6m_2 - 4m_2$$
$$7m_1 = 2m_2$$
$$\frac{m_1}{m_2} = \frac{2}{7}$$

So the ratio is $m_1 : m_2 = 2 : 7$.

Check with the y-coordinates. Put $m_1 = 2$, $m_2 = 7$:

$$\frac{-8(2) + 10(7)}{2 + 7} = \frac{-16 + 70}{9} = \frac{54}{9} = 6,$$

which is the given y-coordinate. So the ratio is correct.

Answer: The point $(-4, 6)$ divides AB internally in the ratio $2 : 7$.

Example 8

Question. Find the coordinates of the points of trisection (i.e., points dividing in three equal parts) of the line segment joining the points $A(2,-2)$ and $B(-7,4)$.

Solution. Let P and Q be the points of trisection, so $AP = PQ = QB$ with P nearer to A. Here $(x_1, y_1) = (2,-2)$ and $(x_2, y_2) = (-7,4)$.

P is one part from A and two parts from B, so it divides AB in the ratio $1:2$. Q is two parts from A and one part from B, so it divides AB in the ratio $2:1$. By the section formula $\left(\dfrac{m_1x_2+m_2x_1}{m_1+m_2}, \dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$:

For P with $m_1 : m_2 = 1:2$:

$$x = \frac{1(-7)+2(2)}{1+2} = \frac{-7+4}{3} = \frac{-3}{3} = -1, \qquad y = \frac{1(4)+2(-2)}{1+2} = \frac{4-4}{3} = \frac{0}{3} = 0.$$

For Q with $m_1 : m_2 = 2:1$:

$$x = \frac{2(-7)+1(2)}{2+1} = \frac{-14+2}{3} = \frac{-12}{3} = -4, \qquad y = \frac{2(4)+1(-2)}{2+1} = \frac{8-2}{3} = \frac{6}{3} = 2.$$

Verification. By the mid-point formula $\left(\dfrac{x_1+x_2}{2}, \dfrac{y_1+y_2}{2}\right)$, the mid-point of PB is

$$\left(\frac{-1+(-7)}{2}, \frac{0+4}{2}\right) = (-4,2) = Q,$$

as it should be because Q is halfway between P and B.

Answer: The points of trisection are $P(-1,0)$ and $Q(-4,2)$.

Example 9

Question. Find the ratio in which the y-axis divides the line segment joining the points $(5,-6)$ and $(-1,-4)$. Also find the point of intersection.

Solution. Let $A(5,-6)$, $B(-1,-4)$, and let the y-axis divide AB in the ratio $k:1$ at the point P. So $(x_1, y_1) = (5,-6)$, $(x_2, y_2) = (-1,-4)$, $m_1 = k$, $m_2 = 1$.

By the section formula,

$$P = \left(\frac{k(-1)+1(5)}{k+1}, \frac{k(-4)+1(-6)}{k+1}\right) = \left(\frac{5-k}{k+1}, \frac{-4k-6}{k+1}\right).$$

A point on the y-axis has x-coordinate 0:

$$\frac{5-k}{k+1} = 0 \;\Rightarrow\; 5-k = 0 \;\Rightarrow\; k = 5.$$

So the ratio is $5:1$. Put $k = 5$ in the y-coordinate:

$$y = \frac{-4(5)-6}{5+1} = \frac{-20-6}{6} = \frac{-26}{6} = -\frac{13}{3}.$$

Common mistake. *On the y-axis it is the x-coordinate that is 0. Setting the y-coordinate to 0 here would give a different, wrong ratio.*

Answer: The y-axis divides the segment in the ratio $5:1$, and the point of intersection is $\left(0, -\dfrac{13}{3}\right)$.

Example 10

Question. If the points $A(6,1)$, $B(8,2)$, $C(9,4)$ and $D(p,3)$ are the vertices of a parallelogram, taken in order, find the value of p.

Solution. The diagonals of a parallelogram bisect each other, so the mid-point of AC equals the mid-point of BD.

By the mid-point formula $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$:

$$\text{Mid-point of } AC = \left(\frac{6+9}{2}, \frac{1+4}{2}\right) = \left(\frac{15}{2}, \frac{5}{2}\right),$$
$$\text{Mid-point of } BD = \left(\frac{8+p}{2}, \frac{2+3}{2}\right) = \left(\frac{8+p}{2}, \frac{5}{2}\right).$$

Equating the x-coordinates: $\frac{15}{2} = \frac{8+p}{2}$, so $8+p = 15$ and $p = 7$. The y-coordinates are both $\frac{5}{2}$, which is consistent.

Verification. With $D(7,3)$, opposite sides should be equal. By the distance formula $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$:

$$AB^2 = (8-6)^2 + (2-1)^2 = 2^2 + 1^2 = 4 + 1 = 5$$
$$DC^2 = (9-7)^2 + (4-3)^2 = 2^2 + 1^2 = 4 + 1 = 5$$
$$BC^2 = (9-8)^2 + (4-2)^2 = 1^2 + 2^2 = 1 + 4 = 5$$
$$AD^2 = (7-6)^2 + (3-1)^2 = 1^2 + 2^2 = 1 + 4 = 5$$

So $AB = DC$ and $BC = AD$: opposite sides are equal, as in a parallelogram.

Answer: $p = 7$.

Part D: Exercise 7.2

Exercise 7.2, Question 1

Question. Find the coordinates of the point which divides the join of $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$.

Solution. Let $A(-1, 7)$, $B(4, -3)$ and $P(x, y)$ divide AB internally in the ratio $m_1 : m_2 = 2 : 3$. So $(x_1, y_1) = (-1, 7)$, $(x_2, y_2) = (4, -3)$, $m_1 = 2$, $m_2 = 3$.

By the section formula $\left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$:

$$x = \frac{2(4) + 3(-1)}{2 + 3} = \frac{8 - 3}{5} = \frac{5}{5} = 1$$
$$y = \frac{2(-3) + 3(7)}{2 + 3} = \frac{-6 + 21}{5} = \frac{15}{5} = 3$$

Common mistake. *m_1 goes with the far point B (x_2, y_2) and m_2 with A (x_1, y_1). Writing $2(-1) + 3(4)$ instead would divide in the ratio $3 : 2$.*

Answer: The required point is $(1, 3)$.

Exercise 7.2, Question 2

Question. Find the coordinates of the points of trisection of the line segment joining $(4, -1)$ and $(-2, -3)$.

Solution. Let $A(4, -1)$, $B(-2, -3)$. The points of trisection P and Q divide AB into three equal parts, with P nearer to A. So P divides AB in the ratio $1 : 2$ and Q in the ratio $2 : 1$.

By the section formula $\left(\dfrac{m_1x_2 + m_2x_1}{m_1 + m_2}, \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$ with $(x_1, y_1) = (4, -1)$, $(x_2, y_2) = (-2, -3)$:

For P with $m_1 : m_2 = 1 : 2$:

$$x = \frac{1(-2) + 2(4)}{1 + 2} = \frac{-2 + 8}{3} = \frac{6}{3} = 2, \qquad y = \frac{1(-3) + 2(-1)}{1 + 2} = \frac{-3 - 2}{3} = -\frac{5}{3}.$$

For Q with $m_1 : m_2 = 2 : 1$:

$$x = \frac{2(-2) + 1(4)}{2 + 1} = \frac{-4 + 4}{3} = 0, \qquad y = \frac{2(-3) + 1(-1)}{2 + 1} = \frac{-6 - 1}{3} = -\frac{7}{3}.$$

Verification. By the mid-point formula, Q should be the mid-point of PB:

$$\left(\frac{2 + (-2)}{2}, \frac{-\frac{5}{3} + (-3)}{2}\right) = \left(0, \frac{-\frac{14}{3}}{2}\right) = \left(0, -\frac{7}{3}\right) = Q.$$

Answer: The points of trisection are $P\left(2, -\dfrac{5}{3}\right)$ and $Q\left(0, -\dfrac{7}{3}\right)$.

Exercise 7.2, Question 3

Question. To conduct Sports Day activities, in your rectangular shaped school ground $ABCD$, lines have been drawn with chalk powder at a distance of 1m each. 100 flower pots have been placed at a distance of 1m from each other along AD, as shown in Fig. 7.12. Niharika runs $\frac{1}{4}$th the distance AD on the 2nd line and posts a green flag. Preet runs $\frac{1}{5}$th the distance AD on the eighth line and posts a red flag. What is the distance between both the flags? If Rashmi has to post a blue flag exactly halfway between the line segment joining the two flags, where should she post her flag?

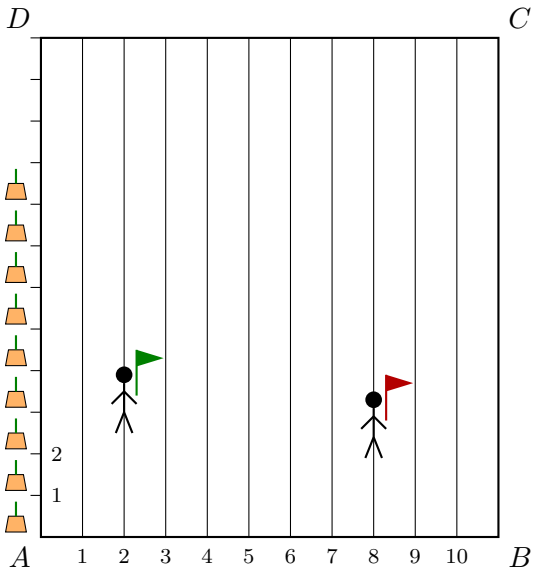


Fig. 7.12 (redrawn, not to scale). Chalk lines numbered 1 to 10 from A along AB; flower pots 1 m apart along AD (only some are drawn); green flag on line 2, red flag on line 8.

Solution.

Step 1: length of AD. 100 flower pots are placed 1 m apart along AD, so $AD = 100$ m (this is the value the book intends).

Step 2: set up coordinates. Take A as the origin, AB along the x-axis and AD along the y-axis, with 1 unit = 1 m. The chalk lines are 1 m apart and numbered from A, so the nth line is the line $x = n$. A flag on the nth line therefore has x-coordinate n, and its y-coordinate is its distance from AB measured along the line.

Common mistake. *"The 2nd line" gives the x-coordinate ($x = 2$), not the y-coordinate. The fraction of AD gives the y-coordinate.*

Step 3: position of each flag.
Green flag: on the 2nd line, so $x = 2$. Niharika runs $\frac{1}{4}$ of AD, so $y = \frac{1}{4} \times 100 = 25$. So $G(2, 25)$.
Red flag: on the 8th line, so $x = 8$. Preet runs $\frac{1}{5}$ of AD, so $y = \frac{1}{5} \times 100 = 20$. So $R(8, 20)$.

Step 4: distance between the flags. By the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ with $(x_1, y_1) = (2, 25)$ and $(x_2, y_2) = (8, 20)$:

$$\begin{aligned} GR &= \sqrt{(8-2)^2 + (20-25)^2} \\ &= \sqrt{6^2 + (-5)^2} \\ &= \sqrt{36 + 25} = \sqrt{61} \end{aligned}$$

$\sqrt{61} \approx 7.81$ and 61 has no square factor, so it stays as $\sqrt{61}$.

Step 5: the blue flag. Rashmi posts it halfway between the two flags, that is at the mid-point of

GR. By the mid-point formula $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$:

$$\left(\frac{2+8}{2},\ \frac{25+20}{2}\right) = \left(\frac{10}{2},\ \frac{45}{2}\right) = (5,\ 22.5).$$

Verification. The blue flag $B_1(5, 22.5)$ should be equally far from both flags. Using $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$:

$$B_1G^2 = (2-5)^2 + (25-22.5)^2 = (-3)^2 + (2.5)^2 = 9 + 6.25 = 15.25$$
$$B_1R^2 = (8-5)^2 + (20-22.5)^2 = 3^2 + (-2.5)^2 = 9 + 6.25 = 15.25$$

Each is $15.25 = \frac{61}{4}$, so each distance is $\frac{\sqrt{61}}{2}$, which is half of GR.

Answer: The distance between the two flags is $\sqrt{61}$ m (≈ 7.81 m). Rashmi should post the blue flag at $(5, 22.5)$, that is on the 5th line, 22.5 m from AB along that line.

Exercise 7.2, Question 4

Question. Find the ratio in which the line segment joining the points $(-3, 10)$ and $(6, -8)$ is divided by $(-1, 6)$.

Solution. Let $A(-3, 10)$, $B(6, -8)$ and let $P(-1, 6)$ divide AB in the ratio $m_1 : m_2$. So $(x_1, y_1) = (-3, 10)$ and $(x_2, y_2) = (6, -8)$.

By the section formula, $x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}$:

$$\begin{aligned}
-1 &= \frac{m_1(6) + m_2(-3)}{m_1 + m_2} = \frac{6m_1 - 3m_2}{m_1 + m_2} \\
-(m_1 + m_2) &= 6m_1 - 3m_2 \\
-m_1 - m_2 &= 6m_1 - 3m_2 \\
3m_2 - m_2 &= 6m_1 + m_1 \\
2m_2 &= 7m_1 \\
\frac{m_1}{m_2} &= \frac{2}{7}
\end{aligned}$$

So the ratio is $2 : 7$.

Check with the y-coordinate. Put $m_1 = 2$, $m_2 = 7$ in $y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$:

$$\frac{2(-8) + 7(10)}{2 + 7} = \frac{-16 + 70}{9} = \frac{54}{9} = 6,$$

which is the y-coordinate of P. The ratio is correct.

Answer: The ratio is $2 : 7$.

Exercise 7.2, Question 5

Question. Find the ratio in which the line segment joining $A(1,-5)$ and $B(-4,5)$ is divided by the x-axis. Also find the coordinates of the point of division.

Solution. Let the x-axis divide AB at P in the ratio $k:1$. A point on the x-axis has y-coordinate 0. Here $(x_1,y_1)=(1,-5)$, $(x_2,y_2)=(-4,5)$, $m_1=k$, $m_2=1$.

By the section formula $\left(\dfrac{m_1x_2+m_2x_1}{m_1+m_2},\dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$:

$$P=\left(\frac{k(-4)+1(1)}{k+1},\ \frac{k(5)+1(-5)}{k+1}\right)=\left(\frac{1-4k}{k+1},\ \frac{5k-5}{k+1}\right).$$

The y-coordinate of P is 0:

$$\frac{5k-5}{k+1}=0\ \Rightarrow\ 5k-5=0\ \Rightarrow\ k=1.$$

So the ratio is $1:1$. Put $k=1$ in the x-coordinate:

$$x=\frac{1-4(1)}{1+1}=\frac{-3}{2}=-\frac{3}{2}.$$

Verification. A ratio $1:1$ means P is the mid-point. By the mid-point formula,

$$\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$$

the mid-point of AB is

$$\left(\frac{1+(-4)}{2},\frac{-5+5}{2}\right)=\left(-\frac{3}{2},0\right),$$

which lies on the x-axis.

Answer: The x-axis divides AB in the ratio $1:1$ (at its mid-point), and the point of division is $\left(-\dfrac{3}{2},0\right)$.

Exercise 7.2, Question 6

Question. If $(1,2)$, $(4,y)$, $(x,6)$ and $(3,5)$ are the vertices of a parallelogram taken in order, find x and y.

Solution. Let $A(1,2)$, $B(4,y)$, $C(x,6)$, $D(3,5)$. The diagonals of a parallelogram bisect each other, so the mid-points of the diagonals AC and BD coincide.

By the mid-point formula $\left(\dfrac{x_1+x_2}{2},\dfrac{y_1+y_2}{2}\right)$:

$$\text{Mid-point of } AC=\left(\frac{1+x}{2},\frac{2+6}{2}\right)=\left(\frac{1+x}{2},4\right),$$
$$\text{Mid-point of } BD=\left(\frac{4+3}{2},\frac{y+5}{2}\right)=\left(\frac{7}{2},\frac{y+5}{2}\right).$$

Equate the x-coordinates and the y-coordinates:

$$\frac{1+x}{2}=\frac{7}{2}\ \Rightarrow\ 1+x=7\ \Rightarrow\ x=6,\qquad 4=\frac{y+5}{2}\ \Rightarrow\ y+5=8\ \Rightarrow\ y=3.$$

Verification. With $A(1,2)$, $B(4,3)$, $C(6,6)$, $D(3,5)$, opposite sides should be equal. By $d^2 = (x_2-x_1)^2+(y_2-y_1)^2$:

$$AB^2 = (4-1)^2+(3-2)^2 = 3^2+1^2 = 9+1 = 10$$
$$DC^2 = (6-3)^2+(6-5)^2 = 3^2+1^2 = 9+1 = 10$$
$$BC^2 = (6-4)^2+(6-3)^2 = 2^2+3^2 = 4+9 = 13$$
$$AD^2 = (3-1)^2+(5-2)^2 = 2^2+3^2 = 4+9 = 13$$

So $AB = DC$ and $BC = AD$: opposite sides are equal.

Answer: $x = 6$ and $y = 3$.

Exercise 7.2, Question 7

Question. Find the coordinates of a point A, where AB is the diameter of a circle whose centre is $(2,-3)$ and B is $(1,4)$.

Solution. The centre of a circle is the mid-point of every diameter. Let $A(x,y)$ and $B(1,4)$; the centre $(2,-3)$ is the mid-point of AB.

By the mid-point formula $\left(\dfrac{x_1+x_2}{2}, \dfrac{y_1+y_2}{2}\right)$:

$$\frac{x+1}{2} = 2 \;\Rightarrow\; x+1 = 4 \;\Rightarrow\; x = 3, \qquad \frac{y+4}{2} = -3 \;\Rightarrow\; y+4 = -6 \;\Rightarrow\; y = -10.$$

Verification. The mid-point of $A(3,-10)$ and $B(1,4)$ is $\left(\dfrac{3+1}{2}, \dfrac{-10+4}{2}\right) = \left(\dfrac{4}{2}, \dfrac{-6}{2}\right) = (2,-3)$, the given centre.

Answer: A is $(3,-10)$.

Exercise 7.2, Question 8

Question. If A and B are $(-2,-2)$ and $(2,-4)$, respectively, find the coordinates of P such that $AP = \dfrac{3}{7}AB$ and P lies on the line segment AB.

Solution. Since P lies on segment AB, $PB = AB - AP = AB - \frac{3}{7}AB = \frac{4}{7}AB$. Therefore $AP : PB = \frac{3}{7}AB : \frac{4}{7}AB = 3:4$.

By the section formula $\left(\dfrac{m_1x_2+m_2x_1}{m_1+m_2}, \dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$ with $A(-2,-2)$, $B(2,-4)$ and $m_1 : m_2 = 3:4$:

$$x = \frac{3(2)+4(-2)}{3+4} = \frac{6-8}{7} = -\frac{2}{7}$$
$$y = \frac{3(-4)+4(-2)}{3+4} = \frac{-12-8}{7} = -\frac{20}{7}$$

Common mistake. *$AP = \frac{3}{7}AB$ does not mean the ratio is $3:7$. The ratio $AP : PB$ is $3:4$, because $PB = AB - AP = \frac{4}{7}AB$.*

Verification. By the distance formula $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$:

$$AB = \sqrt{(2-(-2))^2+(-4-(-2))^2} = \sqrt{4^2+(-2)^2} = \sqrt{16+4} = \sqrt{20} = 2\sqrt{5}$$
$$AP = \sqrt{\left(-\tfrac{2}{7}-(-2)\right)^2+\left(-\tfrac{20}{7}-(-2)\right)^2} = \sqrt{\left(\tfrac{12}{7}\right)^2+\left(-\tfrac{6}{7}\right)^2} = \sqrt{\tfrac{144}{49}+\tfrac{36}{49}} = \sqrt{\tfrac{180}{49}} = \tfrac{6\sqrt{5}}{7}$$

and $\frac{3}{7}AB = \frac{3}{7} \times 2\sqrt{5} = \frac{6\sqrt{5}}{7} = AP$.

Answer: $P\left(-\frac{2}{7}, -\frac{20}{7}\right)$.

Exercise 7.2, Question 9

Question. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.

Solution. Let P, Q, R be the points that divide AB into four equal parts, in order from A. Then P divides AB in the ratio $1 : 3$, Q is the mid-point of AB (ratio $2 : 2 = 1 : 1$), and R divides AB in the ratio $3 : 1$. Here $(x_1, y_1) = (-2, 2)$ and $(x_2, y_2) = (2, 8)$.

By the section formula $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$:

P $(m_1 : m_2 = 1 : 3)$: $x = \frac{1(2) + 3(-2)}{1 + 3} = \frac{2 - 6}{4} = -1$, $y = \frac{1(8) + 3(2)}{1 + 3} = \frac{8 + 6}{4} = \frac{14}{4} = \frac{7}{2}$.

Q (mid-point formula $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$): $x = \frac{-2 + 2}{2} = 0$, $y = \frac{2 + 8}{2} = 5$.

R $(m_1 : m_2 = 3 : 1)$: $x = \frac{3(2) + 1(-2)}{3 + 1} = \frac{6 - 2}{4} = 1$, $y = \frac{3(8) + 1(2)}{3 + 1} = \frac{24 + 2}{4} = \frac{26}{4} = \frac{13}{2}$.

Verification. Q should be the mid-point of PR: $\left(\frac{-1 + 1}{2}, \frac{\frac{7}{2} + \frac{13}{2}}{2}\right) = \left(0, \frac{10}{2}\right) = (0, 5) = Q$.

Answer: The points are $P\left(-1, \frac{7}{2}\right)$, $Q(0, 5)$ and $R\left(1, \frac{13}{2}\right)$.

Exercise 7.2, Question 10

Question. Find the area of a rhombus if its vertices are $(3, 0)$, $(4, 5)$, $(-1, 4)$ and $(-2, -1)$ taken in order. [Hint: Area of a rhombus $= \frac{1}{2}$ (product of its diagonals)]

Solution. Let $A(3, 0)$, $B(4, 5)$, $C(-1, 4)$, $D(-2, -1)$. The diagonals are AC and BD. By the distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$:

How to read the working below: in each pair the first point named is (x_1, y_1) and the second is (x_2, y_2). Line 1 puts the values into the formula, line 2 works out each bracket, line 3 squares the brackets, and then the squares are added and the square root is taken.

$$\begin{aligned} AC &= \sqrt{(-1 - 3)^2 + (4 - 0)^2} \\ &= \sqrt{(-4)^2 + 4^2} \\ &= \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2} \\ BD &= \sqrt{(-2 - 4)^2 + (-1 - 5)^2} \\ &= \sqrt{(-6)^2 + (-6)^2} \\ &= \sqrt{36 + 36} = \sqrt{72} = 6\sqrt{2} \end{aligned}$$

(Here $\sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}$ and $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$.)

By the hint, Area $= \frac{1}{2}\, d_1 d_2$ with $d_1 = 4\sqrt{2}$ and $d_2 = 6\sqrt{2}$:

$$\text{Area} = \frac{1}{2} \times 4\sqrt{2} \times 6\sqrt{2} = \frac{1}{2} \times (4 \times 6) \times (\sqrt{2} \times \sqrt{2}) = \frac{1}{2} \times 24 \times 2 = 24.$$

Common mistake. *The area is $\frac{1}{2}d_1d_2$, not d_1d_2. Do not forget the $\frac{1}{2}$.*

Verification that $ABCD$ is a rhombus. All four sides should be equal:

$$\begin{aligned}
AB &= \sqrt{(4-3)^2+(5-0)^2} \\
&= \sqrt{1^2+5^2} \\
&= \sqrt{1+25} = \sqrt{26} \\
BC &= \sqrt{(-1-4)^2+(4-5)^2} \\
&= \sqrt{(-5)^2+(-1)^2} \\
&= \sqrt{25+1} = \sqrt{26} \\
CD &= \sqrt{(-2-(-1))^2+(-1-4)^2} \\
&= \sqrt{(-1)^2+(-5)^2} \\
&= \sqrt{1+25} = \sqrt{26} \\
DA &= \sqrt{(3-(-2))^2+(0-(-1))^2} \\
&= \sqrt{5^2+1^2} \\
&= \sqrt{25+1} = \sqrt{26}
\end{aligned}$$

All four sides are $\sqrt{26}$, so the quadrilateral is a rhombus and the hint applies.

Answer: The area of the rhombus is 24 square units.